Power-Optimal Inverter Chain Design Under Elmore Delay Constraints

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Task Assignments: This project was completed collaboratively, with all members contributing to the writing, revision, and overall development of the report.

- **Suprith Krishnakumar:** Worked on the primal formulation by translating the physical inverter chain into a mathematical framework, and contributed to the related works and references sections.
- **John Fang:** Developed the dual formulation and KKT conditions, and contributed to the logarithmic transformation of the non-convex delay constraint.
- **John Hsu:** Contributed to the primal formulation, including convexity proofs, logarithmic transformations, and analysis of strong duality via Slater's condition, as well as the physical modeling of the inverter chain.
- **Yuheng Li:** Implemented the optimization in CVXPY, conducted simulations, analyzed results, and performed comparisons with a simplified baseline model.

All members contributed equally to discussions, editing, and finalizing the report.

1 Introduction

1.1 Motivation

Modern VLSI digital circuits demand increasingly stringent performance and energy efficiency requirements, particularly as transistor scaling continues to push designs toward higher density and faster clock speeds. A fundamental component in high-speed digital design is the use of inverter chains (buffer chains) to drive large capacitive loads, where improper sizing or staging can lead to excessive delay or unnecessary power consumption. While classical design heuristics such as fixed stage effort or logical effort methods provide useful guidelines, they

do not always yield globally optimal solutions under complex constraints involving wire parasitics, transistor sizing limits, and timing requirements. The goal of this work is to develop a principled optimization framework for inverter chain design that minimizes dynamic power consumption subject to an Elmore delay constraint. We approach this problem through the lens of convex optimization by reformulating the inherently non-convex delay expressions into a geometric programming (GP) framework via logarithmic transformation. This enables efficient computation of globally optimal inverter sizes and interconnect lengths while accounting for physical constraints. Through this formulation, we aim to bridge classical circuit design intuition with modern optimization techniques and provide a scalable method for a energy-efficient buffer chain design.

1.2 Previous Works

1.2.1 Low-Power Circuit Design and Delay Modeling

The demand for high-throughput, low-power digital systems has become increasingly critical, largely driven by the stringent energy requirements of portable computing applications [**cmos**]. Power dissipation in digital CMOS circuits primarily stems from switching (dynamic) power, direct-path short-circuit currents, and subthreshold leakage [**cmos**]. To successfully optimize systems for power, designers rely on mathematical models to predict circuit delay and performance. While transistor behavior in the sub-micrometer regime has been characterized by models such as the alpha-power law [**alpha**], designers frequently abstract interconnect and gate delays using resistor-capacitor (RC) equivalent circuits and the Elmore delay approximation [**gp**]. The Elmore delay provides a reliable measure of signal propagation by calculating the area under the voltage response curve as an RC network discharges, effectively capturing the time it takes for a signal to transition [**gp**].

1.2.2 Classical Sizing Heuristics and Iterative Algorithms

Historically, optimizing the size of logic gates and buffers to minimize delay or power has been approached through classical design heuristics. A prominent example is the method of "logical effort," which simplifies the delay optimization of logic paths and inverter chains into straightforward sizing rules based on RC models [**gp**]. While logical effort is an excellent tool for quickly evaluating candidate circuit topologies and gaining early design intuition, it does not always yield globally optimal solutions under complex constraints involving wire parasitics and strict timing requirements. Other modern optimization approaches tackle simultaneous gate sizing, dual-supply voltage (V_{dd}), and dual-threshold voltage (V_{th}) assignments using sensitivity-based heuristics [**dual˙vdd**]. However, these gradient-descent or heuristic algorithms can be subject to getting trapped in local minima and may struggle to efficiently guarantee a globally optimum power layout [**dual˙vdd**].

1.2.3 Convex Optimization and Geometric Programming in VLSI

To overcome the limitations of heuristic methods, researchers have increasingly turned to convex optimization. The defining advantage of convex optimization is that any local optimum is mathematically guaranteed to be a global optimum, ensuring that solvers do not get trapped in sub-optimal design states [**convex**]. Furthermore, a specialized subset of convex optimization known as Geometric Programming (GP) has emerged as a unifying standard form for circuit sizing problems [**gp**].

The application of GP to digital circuit sizing dates back to 1985 with Fishburn and Dunlop's TILOS algorithm, which optimized transistor sizing using Elmore delay models [**gp**]. Today, highly efficient interior-point algorithms can solve these large-scale GP problems reliably and globally, making it possible to account for complex interactions between area, delay, and power constraints.

Our work builds upon this rich history of geometric programming by applying it specifically to the power-optimal design of inverter (buffer) chains. Instead of relying on traditional iterative heuristics or restricting sizing to gates alone, we treat both the transistor sizing factors (W_i) and the physical interconnect lengths (l_i) between stages as optimization variables.

By analyzing the non-convex multiplicative and ratio terms inherent in full RC Elmore delay expressions, we demonstrate that the delay constraint can be reformulated as a posynomial and mapped into a strictly convex geometric programming space using logarithmic variable substitution. Ultimately, our approach bridges the gap between classical circuit design intuition and modern convex optimization techniques. By comparing our method to a simplified Boyd-style baseline that removes wire resistance and capacitance coupling terms, we verify that utilizing a fully wire-aware GP model prevents designers from underestimating the true power costs required to meet strict timing targets.

1.3 Intended Contributions

1. Optimization-Based Formulation: We formulate the inverter chain design problem as a power minimization problem with Elmore delay, sizing, and interconnect constraints.

2. Geometric Programming Reformulation: We show that the non-convex delay constraint can be expressed as a posynomial and transformed into a convex constraint via logarithmic variable substitution.

3. Convexity Verification: We provide a formal justification that the transformed delay constraint is convex by leveraging the log-sum-exp structure and verifying all terms satisfy geometric programming requirements.

4. Solver Implementation: We implement the resulting formulation in a convex optimization framework (e.g., CVXPY) to compute optimal inverter sizing and interconnect parameters.

5. Design Insight: We analyze the trade-offs between delay, sizing, and power, and compare the optimized solutions to classical inverter chain design heuristics.

1.4 Organization of the Paper

The remainder of this paper is organized as follows. Section II presents the formal statement of the problem, including the physical modeling of the inverter chain, the primal geometric programming formulation, the corresponding dual problem, and the Karush–Kuhn–Tucker (KKT) conditions for optimality. Section III describes the overall approach, including simulation strategy, parameter selection, optimization methodology, and baseline comparisons. Section IV presents the experimental results and discusses key observations regarding wire-aware design, sizing behavior, and model comparisons, followed by concluding remarks. Finally, Section V lists the references used throughout this work.

2 Statement of the Problem

2.1 Primal Formulation

2.1.1 Physical Interpretation of the Inverter Chain

We consider a chain of N CMOS inverters used to drive a fixed load capacitance C_L. Such inverter chains, commonly referred to as buffer chains, are fundamental building blocks in high-speed digital circuits, where they are used to drive large capacitive loads while meeting strict timing constraints.

Each stage in the inverter chain is characterized by two primary design variables:

- $W_i > 0$: the effective transistor width (or sizing factor) of inverter i
- $l_i > 0$: the interconnect length between inverter i and inverter $i+1$

The objective is to determine the optimal values of $\{W_i\}_{i=1}^N$ and $\{l_i\}_{i=1}^N$ that minimize dynamic power consumption while satisfying a delay constraint. Each inverter contributes capacitance from two sources:

Gate (Input) Capacitance The input capacitance of inverter i is proportional to its size:

$$C_{in,i} = W_i C_{g0}$$

where C_{g0} represents the gate capacitance of a unit-sized inverter.

Parasitic Capacitance Each inverter also has intrinsic parasitic capacitance due to diffusion regions:

$$C_{p,i} = W_i C_{p0}$$

where C_{p0} is the parasitic capacitance of a unit inverter.

These capacitances contribute to both dynamic power consumption and delay. The wires connecting inverter stages introduce additional parasitics:

Interconnect Modeling

- Resistance:
$$R_{wire,i} = r\, l_i$$
- Capacitance:
$$C_{wire,i} = c\, l_i$$

where r and c denote per-unit-length resistance and capacitance, respectively. These parameters capture the physical effects of interconnect scaling and play a critical role in delay modeling.

Inverter Resistance The effective output resistance of inverter i is inversely proportional to its size:
$$R_i = \frac{R_0}{W_i}$$
where R_0 is the resistance of a unit inverter. Increasing the transistor width reduces resistance, improving delay at the expense of increased capacitance.

Dynamic Power Objective The dynamic power consumption of a digital circuit is given by:
$$P_{dyn} = \alpha f V_{DD}^2 C_{tot}$$
where α is the switching activity factor, f is the clock frequency, and V_{DD} is the supply voltage. These quantities are treated as constants in our formulation.

Thus, minimizing dynamic power reduces to minimizing the total switched capacitance C_{tot}.

The total capacitance in the inverter chain consists of:

- Parasitic capacitance of each inverter
- Gate capacitance of downstream inverters
- Interconnect capacitance
- Final load capacitance C_L

This yields:
$$C_{tot} = \sum_{i=1}^{N} W_i C_{p0} + \sum_{i=1}^{N-1} W_{i+1} C_{g0} + \sum_{i=1}^{N} c\, l_i + C_L$$
Since C_L is constant, it does not affect the optimization and may be omitted from the objective.

Elmore Delay Constraint To ensure correct circuit operation, the total propagation delay must not exceed a specified clock period T_{clk}. We model delay using the Elmore delay approximation, which expresses delay as a weighted sum of downstream capacitances. The delay contribution of each stage includes interactions between inverter resistance and downstream capacitances, as well as interconnect effects.

For stages $i = 1, \ldots, N-1$, the delay contribution is:

$$d_i = R_0 C_{g0} \frac{W_{i+1}}{W_i} + R_0 C_{p0} + R_0 c \frac{l_i}{W_i} + r C_{g0} l_i W_{i+1} + r C_{p0} l_i W_i + r c l_i^2$$

The final stage differs since it drives the external load C_L:

$$d_N = R_0 C_L \frac{1}{W_N} + R_0 C_{p0} + R_0 c \frac{l_N}{W_N} + r C_L l_N + r C_{p0} l_N W_N + r c l_N^2$$

The total delay constraint is therefore:

$$\sum_{i=1}^{N-1} d_i + d_N \leq T_{clk}$$

Design Constraints To ensure physically realizable designs, we impose:

$$W_{min} \leq W_i \leq W_{max}, \quad W_i > 0, \quad l_i > 0$$

2.1.2 Primal Statement

Geometric Programming Reformulation The original formulation is non-convex due to multiplicative and ratio terms such as:

$$\frac{W_{i+1}}{W_i}, \quad W_i l_i, \quad l_i^2$$

To enable efficient optimization, we reformulate the problem using geometric programming techniques.

Logarithmic Variable Transformation We introduce the change of variables:

$$x_i = \log W_i, \qquad y_i = \log l_i$$

which implies:

$$W_i = e^{x_i}, \qquad l_i = e^{y_i}$$

This transformation is bijective over the positive orthant, ensuring no loss of generality.

Transformation of Nonlinear Terms Under this transformation, multiplicative and ratio terms become exponentials of affine expressions:

$$\frac{W_{i+1}}{W_i} = e^{x_{i+1}-x_i}, \quad W_i l_i = e^{x_i+y_i}, \quad l_i^2 = e^{2y_i}$$

Thus, each term in the delay expressions can be written in the form:

$$c_k e^{a_k^T z}, \quad \text{where } z = [x_1, \ldots, x_N, y_1, \ldots, y_N]^T$$

Convexity of the Log-Sum-Exp Function Define:

$$f(x) = \log\left(\sum_{k=1}^{n} e^{x_k}\right)$$

Let $z_k = e^{x_k}$ and $z = [z_1, \ldots, z_n]^T$. Then:

$$\nabla^2 f(x) = \frac{1}{\mathbf{1}^T z}\mathrm{diag}(z) - \frac{1}{(\mathbf{1}^T z)^2} z z^T$$

For any $v \in \mathbb{R}^n$:

$$v^T \nabla^2 f(x) v = \frac{1}{(\mathbf{1}^T z)^2}\left[(\mathbf{1}^T z)\left(\sum_{i=1}^{n} z_i v_i^2\right) - \left(\sum_{i=1}^{n} z_i v_i\right)^2\right]$$

By the Cauchy–Schwarz inequality:

$$\left(\sum_{i=1}^{n} z_i v_i\right)^2 \leq (\mathbf{1}^T z)\left(\sum_{i=1}^{n} z_i v_i^2\right)$$

Thus:

$$v^T \nabla^2 f(x) v \geq 0 \quad \forall v$$

which proves that $f(x)$ is convex.

Convexity of the Delay Constraint The total delay can be written as:

$$D(x, y) = \sum_k c_k e^{a_k^T z}$$

Taking logarithms:

$$\log D(x, y) = \log\left(\sum_k e^{a_k^T z + \log c_k}\right)$$

which is a log-sum-exp function and therefore convex. Hence, the constraint:

$$D(x, y) \leq T_{clk}$$

is equivalent to the convex constraint:

$$\log D(x, y) \leq \log T_{clk}$$

Final Convex Optimization Problem The optimization problem becomes:

$$\min_{x,y} \quad \log\left(C_{p0}e^{x_1} + \sum_{i=2}^{N}(C_{g0} + C_{p0})e^{x_i} + c\sum_{i=1}^{N} e^{y_i}\right) \tag{1}$$

$$\text{s.t.} \quad \log\left(\sum_{i=1}^{N-1} d_i(x,y) + d_N(x,y)\right) \leq \log T_{clk} \tag{2}$$

$$\log W_{min} \leq x_i \leq \log W_{max}, \quad \forall i$$

$$\log \epsilon \leq y_i \leq \log l_{max}, \quad \forall i$$

Strong Duality and Slater's Condition The problem is convex since:

- The objective is a log-sum-exp function, hence convex.
- The delay constraint is convex.
- The bound constraints are affine.

Let the feasible set be:

$$\mathcal{D} = \{(x,y) \mid \log W_{min} \leq x_i \leq \log W_{max},\ \log \epsilon \leq y_i \leq \log l_{max}\}$$

The relative interior of $\mathcal{D}$ is:

$$\text{relint}(\mathcal{D}) = \{(x,y) \mid \log W_{min} < x_i < \log W_{max},\ \log \epsilon < y_i < \log l_{max}\}$$

To verify Slater's condition, we must show that there exists $(x^*, y^*) \in \text{relint}(\mathcal{D})$ such that:

$$\log\left(\sum_{i=1}^{N-1} d_i(x^*,y^*) + d_N(x^*,y^*)\right) < \log T_{clk}$$

Such a point exists by choosing:

- y_i^* sufficiently small (close to $\log \epsilon$), which reduces all terms involving e^{y_i},
- x_i^* strictly within bounds and arranged to balance exponential terms.

Since all delay terms are positive and continuous, and can be made sufficiently small relative to T_{clk}, a strictly feasible point exists.

Therefore, Slater's condition holds:

$$\exists (x^*, y^*) \in \text{relint}(\mathcal{D})$$

such that

$$f_i(x^*, y^*) < 0$$

Conclusion Since the problem is convex and Slater's condition holds, strong duality follows:

$$p^* - d^* = 0$$

Thus, the duality gap is zero, and the Karush-Kuhn-Tucker (KKT) conditions are necessary and sufficient for optimality.

2.2 Dual Formulation

Define

$$S_0(x, y) = C_{p0}e^{x_1} + \sum_{i=1}^{N}(C_{g0} + C_{p0})e^{x_i} + c\sum_{i=1}^{N} e^{y_i},$$

and

$$S_1(x, y) = \sum_{i=1}^{N-1} d_i(x, y) + d_N(x, y),$$

where, for $i = 1, \ldots, N-1$
The Lagrangian is

$$\begin{aligned}\mathcal{L}(x, y, \lambda, \alpha, \beta, \gamma, \delta) = {} & \log S_0(x, y) + \lambda\big(\log S_1(x, y) - \log T_{clk}\big) \\ & + \sum_{i=1}^{N} \alpha_i(\log W_{\min} - x_i) + \sum_{i=1}^{N} \beta_i(x_i - \log W_{\max}) \\ & + \sum_{i=1}^{N} \gamma_i(\log \varepsilon - y_i) + \sum_{i=1}^{N} \delta_i(y_i - \log l_{\max}).\end{aligned} \tag{3}$$

Using the conjugate of log-sum-exp

$$\log\left(\sum_j e^{s_j}\right) = \max_{\omega_j \geq 0,\ \sum_j \omega_j = 1} \left(\sum_j \omega_j s_j - \sum_j \omega_j \log \omega_j\right),$$

introduce nonnegative weights $\theta_0, \theta_i, \phi_i$ such that

$$\theta_0 + \sum_{i=1}^{N} \theta_i + \sum_{i=1}^{N} \phi_i = 1.$$

Then

$$\begin{aligned}\log S_0(x, y) = \max_{\theta, \phi} \Bigg\{ & \theta_0(\log C_{p0} + x_1) + \sum_{i=1}^{N} \theta_i\big(\log(C_{g0} + C_{p0}) + x_i\big) \\ & + \sum_{i=1}^{N} \phi_i(\log c + y_i) - \theta_0 \log \theta_0 - \sum_{i=1}^{N} \theta_i \log \theta_i - \sum_{i=1}^{N} \phi_i \log \phi_i \Bigg\}.\end{aligned} \tag{4}$$

Using the scaled conjugate log-sum-exp

$$\lambda \log\left(\sum_j e^{s_j}\right) = \max_{\mu_j \geq 0,\ \sum_j \mu_j = \lambda} \left(\sum_j \mu_j s_j - \sum_j \mu_j \log \mu_j + \lambda \log \lambda\right),$$

introduce nonnegative variables $\eta_{i,k}$, $k = 1, \ldots, 6$, such that

$$\sum_{i=1}^{N} \sum_{k=1}^{6} \eta_{i,k} = \lambda.$$

For $i = 1, \ldots, N-1$, let

$$\eta_{i,1} \leftrightarrow R_0 C_{g0} e^{x_{i+1} - x_i}, \quad \eta_{i,2} \leftrightarrow R_0 C_{p0}, \quad \eta_{i,3} \leftrightarrow R_0 c e^{y_i - x_i},$$

$$\eta_{i,4} \leftrightarrow r C_{g0} e^{y_i + x_{i+1}}, \quad \eta_{i,5} \leftrightarrow r C_{p0} e^{y_i + x_i}, \quad \eta_{i,6} \leftrightarrow r c e^{2y_i},$$

and for $i = N$,

$$\eta_{N,1} \leftrightarrow R_0 C_L e^{-x_N}, \quad \eta_{N,2} \leftrightarrow R_0 C_{p0}, \quad \eta_{N,3} \leftrightarrow R_0 c e^{y_N - x_N},$$

$$\eta_{N,4} \leftrightarrow r C_L e^{y_N}, \quad \eta_{N,5} \leftrightarrow r C_{p0} e^{y_N + x_N}, \quad \eta_{N,6} \leftrightarrow r c e^{2y_N}.$$

Then

$$\begin{aligned} \lambda \log S_1(x,y) = \max_{\eta \geq 0,\ \sum \eta = \lambda} \Bigg\{ & \sum_{i=1}^{N-1} \sum_{k=1}^{6} \eta_{i,k} \big(\log a_{i,k} + m_{i,k}(x,y)\big) \\ & + \sum_{k=1}^{6} \eta_{N,k} \big(\log a_{N,k} + m_{N,k}(x,y)\big) - \sum_{i=1}^{N} \sum_{k=1}^{6} \eta_{i,k} \log \eta_{i,k} + \lambda \log \lambda \Bigg\}. \end{aligned} \tag{5}$$

Substituting (4) and (5) into (3), the resulting expression is affine in x and y. Finite infimum over x and y requires the coefficients of each x_i and y_i to vanish. From x_1,

$$\theta_0 + \theta_1 - \eta_{1,1} - \eta_{1,3} + \eta_{1,5} - \alpha_1 + \beta_1 = 0.$$

From x_i, $i = 2, \ldots, N-1$,

$$\theta_i + \eta_{i-1,1} + \eta_{i-1,4} - \eta_{i,1} - \eta_{i,3} + \eta_{i,5} - \alpha_i + \beta_i = 0.$$

From x_N,

$$\theta_N + \eta_{N-1,1} + \eta_{N-1,4} - \eta_{N,1} - \eta_{N,3} + \eta_{N,5} - \alpha_N + \beta_N = 0.$$

From y_i, $i = 1, \ldots, N$,

$$\phi_i + \eta_{i,3} + \eta_{i,4} + \eta_{i,5} + 2\eta_{i,6} - \gamma_i + \delta_i = 0.$$

Then the dual function is

$$
\begin{aligned}
g = {} & \theta_0 \log \frac{C_{p0}}{\theta_0} + \sum_{i=1}^{N} \theta_i \log \frac{C_{g0} + C_{p0}}{\theta_i} + \sum_{i=1}^{N} \phi_i \log \frac{c}{\phi_i} \\
& + \sum_{i=1}^{N-1} \left[\eta_{i,1} \log \frac{R_0 C_{g0}}{\eta_{i,1}} + \eta_{i,2} \log \frac{R_0 C_{p0}}{\eta_{i,2}} + \eta_{i,3} \log \frac{R_0 c}{\eta_{i,3}} + \eta_{i,4} \log \frac{r C_{g0}}{\eta_{i,4}} + \eta_{i,5} \log \frac{r C_{p0}}{\eta_{i,5}} + \eta_{i,6} \log \frac{rc}{\eta_{i,6}} \right] \\
& + \eta_{N,1} \log \frac{R_0 C_L}{\eta_{N,1}} + \eta_{N,2} \log \frac{R_0 C_{p0}}{\eta_{N,2}} + \eta_{N,3} \log \frac{R_0 c}{\eta_{N,3}} + \eta_{N,4} \log \frac{r C_L}{\eta_{N,4}} + \eta_{N,5} \log \frac{r C_{p0}}{\eta_{N,5}} + \eta_{N,6} \log \frac{rc}{\eta_{N,6}} \\
& + \lambda \log \lambda - \lambda \log T_{clk} + \sum_{i=1}^{N} \alpha_i \log W_{\min} - \sum_{i=1}^{N} \beta_i \log W_{\max} + \sum_{i=1}^{N} \gamma_i \log \varepsilon - \sum_{i=1}^{N} \delta_i \log l_{\max}.
\end{aligned}
$$

Therefore, the dual problem is

$$
\begin{aligned}
\max_{\theta,\phi,\eta,\lambda,\alpha,\beta,\gamma,\delta} \quad & g \\
\text{s.t.} \quad & \theta_0 \geq 0, \qquad \theta_i \geq 0, \qquad \phi_i \geq 0, \qquad i = 1, \ldots, N, \\
& \eta_{i,k} \geq 0, \qquad i = 1, \ldots, N, \quad k = 1, \ldots, 6, \\
& \alpha_i, \beta_i, \gamma_i, \delta_i \geq 0, \qquad i = 1, \ldots, N, \\
& \theta_0 + \sum_{i=1}^{N} \theta_i + \sum_{i=1}^{N} \phi_i = 1, \\
& \sum_{i=1}^{N} \sum_{k=1}^{6} \eta_{i,k} = \lambda, \\
& \theta_0 + \theta_1 - \eta_{1,1} - \eta_{1,3} + \eta_{1,5} - \alpha_1 + \beta_1 = 0, \\
& \theta_i + \eta_{i-1,1} + \eta_{i-1,4} - \eta_{i,1} - \eta_{i,3} + \eta_{i,5} - \alpha_i + \beta_i = 0, \quad i = 2, \ldots, N-1, \\
& \theta_N + \eta_{N-1,1} + \eta_{N-1,4} - \eta_{N,1} - \eta_{N,3} + \eta_{N,5} - \alpha_N + \beta_N = 0, \\
& \phi_i + \eta_{i,3} + \eta_{i,4} + \eta_{i,5} + 2\eta_{i,6} - \gamma_i + \delta_i = 0, \quad i = 1, \ldots, N.
\end{aligned}
$$

2.3 KKT Conditions

- Primal feasibility:

$$
\begin{aligned}
& \log S_1(x, y) \leq \log T_{clk} \\
& \log W_{\min} \leq x_i \leq \log W_{\max}, \forall i \\
& \log \epsilon \leq y_i \leq \log l_{\max}, \forall i
\end{aligned}
$$

- Dual feasibility:

$$
\lambda \geq 0, \quad \alpha_i \geq 0, \quad \beta_i \geq 0, \quad \gamma_i \geq 0, \quad \delta_i \geq 0, \quad \forall i
$$

- Complementary slackness:

$$\begin{aligned}
&\lambda(\log S_1(x,y) - \log T_{clk}) = 0 \\
&\alpha_i(\log W_{\min} - x_i) = 0, \forall i \\
&\beta_i(x_i - \log W_{\max}) = 0, \forall i \\
&\gamma_i(\log \epsilon - y_i) = 0, \forall i \\
&\delta_i(y_i - \log l_{\max}) = 0, \forall i
\end{aligned}$$

- Stationarity:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial x_i} &= \frac{\partial}{\partial x_i} \log S_0(x,y) + \lambda \frac{\partial}{\partial x_i}(\log S_1(x,y) - \log T_{clk}) - \alpha_i + \beta_i \\
&= \frac{C_{p0}e^{x_1} + (C_{g0} + C_{p0})e^{x_i}}{S_0(x,y)} \\
&+ \frac{\lambda(R_0 C_{g0} e^{x_i - x_{i-1}} + rC_{go} e^{y_{i-1}+x_i} - R_0 C_{g0} e^{x_{i+1}-x_i} - R_0 cE^{y_i - x_i} + rC_{p0} e^{y_i + x_i})}{S_1(x,y)} - \alpha_i + \beta_i = 0 \\
\frac{\partial \mathcal{L}}{\partial y_i} &= \frac{\partial}{\partial y_i} \log S_0(x,y) + \lambda \frac{\partial}{\partial y_i}(\log S_1(x,y) - \log T_{clk}) - \gamma_i + \delta_i \\
&= \frac{ce^{y_i}}{S_0(x,y)} + \frac{\lambda(R_0 ce^{y_i - x_i} + rC_{g0} e^{y_i + x_{i+1}} + rC_{p0} e^{y_i + x_i} + 2rce^{2y_i})}{S_1(x,y)} - \gamma_i + \delta_i = 0
\end{aligned}$$

3 Approach

3.1 Simulation Strategy and Design Choices

Although the theoretical formulation developed in the previous sections treats both inverter widths $\{W_i\}$ and wire lengths $\{l_i\}$ as optimization variables, a direct implementation of this fully unconstrained problem leads to degenerate solutions in practice. In particular, since both capacitance and delay decrease with smaller wire lengths, the optimizer tends to drive l_i to their minimum allowable values, resulting in physically uninformative solutions.

To address this issue and obtain meaningful design insights, we adopt a two-stage experimental strategy:

1. **Fixed-profile wire simulations:** We fix the wire lengths $\{l_i\}$ according to predefined profiles and optimize only the inverter widths $\{W_i\}$. This allows us to isolate the effect of wire distribution on optimal sizing while avoiding degeneracy.

2. **Unrestricted optimization:** We subsequently solve the full problem where both $\{W_i\}$ and $\{l_i\}$ are variables. This experiment is used primarily to validate the theoretical formulation and to study primal-dual consistency.

This separation enables both interpretability and theoretical validation within the same framework.

3.2 Parameter Selection and Operating Regime

To ensure that the optimization produces nontrivial and informative solutions, we carefully selected a parameter regime that avoids boundary-dominated behavior. In particular, we chose:

$$N = 6, \quad \sum_{i=1}^{N} l_i = 25, \quad T_{clk} = 90, \quad C_L = 15$$

This regime was empirically tuned to ensure that:

- The delay constraint is active but not overly restrictive
- The optimal widths are not trivially pinned at W_{min}
- Wire effects meaningfully influence the solution

This allows us to observe the interaction between sizing, interconnect distribution, and timing constraints.

3.3 Wire Length Profiles

In the fixed-profile setting, we define several representative wire distributions to study how interconnect placement affects optimal design:

- **Uniform profile:** Wire length is evenly distributed across all stages
- **Front-loaded profile:** Larger wire segments appear near the input stages
- **Back-loaded profile:** Larger wire segments appear near the output stages

These profiles are designed to reflect different physical layout scenarios and allow us to study how downstream loading influences optimal inverter sizing.

3.4 Optimization Methodology

The optimization problem is implemented using the geometric programming formulation derived earlier. Specifically:

- Decision variables are transformed into the log-domain:

$$x_i = \log W_i, \quad y_i = \log l_i$$

- The delay constraint is expressed as a log-sum-exp function, ensuring convexity
- The objective function remains a sum of exponentials, which is compatible with convex solvers

The resulting convex problem is solved using **CVXPY**, which provides a high-level interface for specifying and solving convex optimization problems. This allows us to efficiently compute globally optimal solutions for each experimental configuration.

3.5 Baseline Comparison

To evaluate the importance of modeling interconnect effects, we compare the full wire-aware model with a simplified baseline inspired by classical formulations (e.g., logical effort or gate-only models). In this baseline:

- Wire resistance and capacitance terms are removed
- Only gate and parasitic capacitances are considered

This comparison highlights the extent to which neglecting interconnect effects underestimates the cost of meeting timing constraints.

3.6 Unrestricted Formulation and Dual Validation

In addition to the fixed-profile experiments, we also solve the unrestricted formulation where both $\{W_i\}$ and $\{l_i\}$ are optimized simultaneously. While this formulation is less physically informative due to degeneracy, it serves two important purposes:

- **Verification of theoretical results:** We compare the primal and dual objective values to confirm strong duality.
- **Numerical consistency:** We evaluate the duality gap in the log-domain to ensure correctness of the convex reformulation.

This step provides additional confidence in the validity of the geometric programming approach.

3.7 Evaluation Metrics

Across all experiments, we evaluate:

- **Optimal objective value:** Represents the minimized capacitance (and thus dynamic power)
- **Timing ratio:** Ratio of achieved delay to T_{clk}, indicating constraint tightness
- **Width allocation:** Stage-wise inverter sizing patterns
- **Sensitivity to parameters:** Changes in objective under varying T_{clk} and wire profiles

These metrics allow us to systematically analyze both quantitative performance and qualitative design behavior.

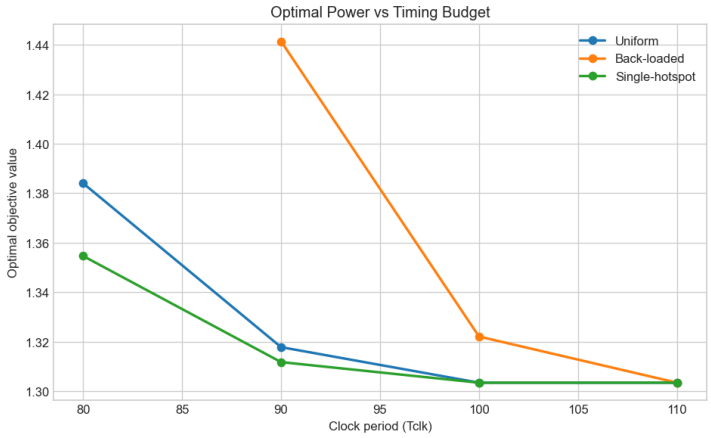

Figure 1: Optimal objective versus timing budget for the fixed-profile wire-aware model. Tighter timing constraints increase the objective across all tested wire profiles, and the back-loaded profile is generally the most expensive.

4 Results And Conclusion

4.1 Results

Although the theoretical development in the previous sections treats both inverter widths and wire lengths as optimization variables, our simulation study begins with a fixed-profile setting for the wire lengths. This choice avoids the potentially degenerate behavior of the unrestricted formulation and yields more interpretable design trade-offs under wire-aware timing constraints. After presenting these main experiments, we return to the unrestricted formulation to align the numerical study with the earlier primal-dual analysis.

We first selected a feasible and informative operating regime so that the optimizer would not trivially pin all gate widths at their lower bounds. The selected setting was $N = 6$, total wire length $= 25$, $T_{clk} = 90$, and $C_L = 15$. Under this tuned regime, the fixed-profile simulation produced nontrivial width allocations, which allowed us to observe how timing pressure and wire distribution affect the optimal design.

Figure 1 shows the timing-budget sweep for the fixed-profile model. As the clock period T_{clk} becomes tighter, the optimal objective increases across all tested wire profiles. This trend is consistent with the model structure: a stricter Elmore delay constraint forces more aggressive sizing, which raises the capacitance-related power objective. Among the tested profiles, the back-loaded interconnect profile is generally the most expensive, reflecting the stronger downstream delay penalty created by longer wires near the output end of the chain.

Figure 2 shows the stage-by-stage width allocation for the three wire profiles in the representative operating regime. In the back-loaded case, the optimized

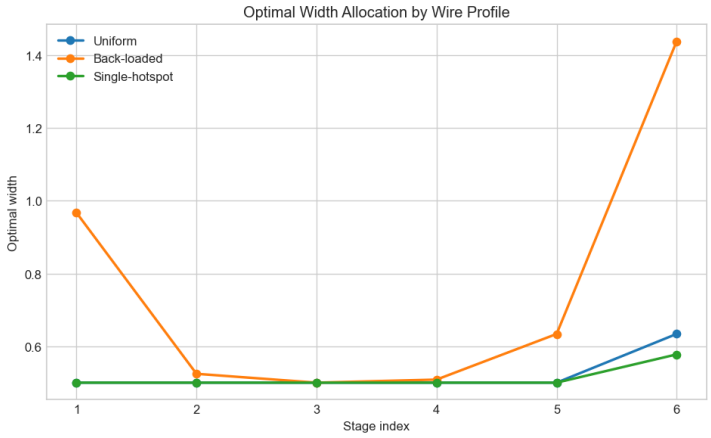

Figure 2: Stage-by-stage optimal width allocation under three fixed wire profiles in the tuned operating regime. Concentrating more wire length near later stages changes the sizing pattern and tends to enlarge gates closer to the output side of the chain.

objective was approximately 1.4414, with a timing ratio of 1.00000049, and the resulting widths were

$$[0.9673,\ 0.5243,\ 0.5000,\ 0.5084,\ 0.6338,\ 1.4383].$$

These values are clearly nonuniform, and the larger widths near the later stages indicate that the optimizer compensates for heavier downstream loading and larger wire-delay contributions. This provides a direct qualitative explanation for why a wire-aware model can produce different sizing recommendations than a gate-only approximation.

To benchmark the wire-aware model, we compared it with a Boyd-style simplified baseline that removes wire resistance and wire-capacitance coupling terms. As shown in Figure 3, the full model consistently produced a higher objective value than the simplified baseline under matched settings. This suggests that neglecting interconnect effects can underestimate the design cost required to satisfy a timing target.

Finally, we returned to the unrestricted formulation in which both W_i and l_i are optimized. Figure 4 shows that, in the tested regime, the unrestricted primal drives both the inverter widths and the wire lengths to their lower bounds. This confirms the degeneracy concern that motivated the fixed-profile setup in the main simulation. At the same time, the unrestricted primal objective was 1.0576163, the dual-implied objective was 1.0576159, and the log-domain duality gap was approximately 3.94×10^{-7}. This provides numerical support for the dual formulation while also showing that the unrestricted problem, although mathematically consistent, is less physically informative for simulation purposes.

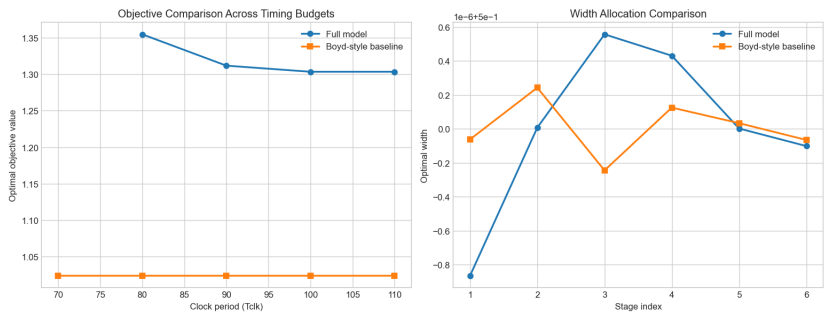

Figure 3: Comparison between the full wire-aware model and a Boyd-style simplified baseline. The full model consistently yields a higher objective, indicating that neglecting wire effects underestimates the cost of meeting a timing target.

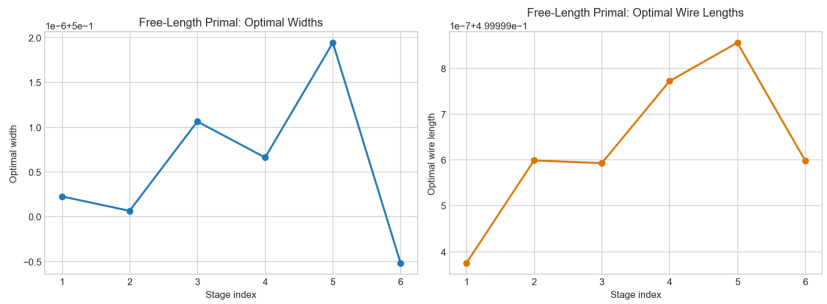

Figure 4: Optimal widths and wire lengths in the unrestricted formulation where both W_i and l_i are optimized. In the tested regime, both variables collapse to their lower bounds, revealing the degeneracy of the free-length formulation.

4.2 Conclusion

5 References